

SAMEdu

Educational Social Media Platform

Systems Analysis & Design Documentation

Anna Sargsyan

Webster University

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About SAMEdu Platform

SAMEdu is an educational social media platform designed to promote knowledge sharing within a trustworthy community of professionals and learners. The platform addresses a common challenge of traditional social media: users can share information on specialized topics regardless of their qualifications or expertise, making it difficult to distinguish reliable information from misinformation.

SAMEdu approaches this problem through professional verification and content review. Users who wish to share professional or educational content provide information about their educational background and professional experience as part of the account verification process. Once verified, professionals can contribute content within their recognized areas of expertise. Submitted content undergoes a review process before publication to help ensure that the information shared on the platform is relevant and appropriate.

The platform also serves learners who want to discover educational content without necessarily contributing professional material themselves. By combining familiar social media interactions with structured educational content, SAMEdu is designed to make learning accessible through everyday activities such as browsing and scrolling.

The overall goal of SAMEdu is to create an environment where professionals can share their knowledge, learners can discover reliable educational content, and users can connect around common academic and professional interests.

Sub-Systems

SAMEdu is organized into two primary sub-systems that support the platform's core functionality.

User Authentication Sub-system

The User Authentication sub-system manages user registration, login, account security, and identity verification. It supports the verification of users' educational and professional backgrounds, which determines their ability to contribute content within specific areas of expertise.

Content Management Sub-system

The Content Management sub-system manages the creation, submission, review, approval, and organization of educational content. It supports the platform's content review process and helps ensure that published material is associated with appropriate areas of expertise.

System Capabilities

Profile Creation: Users can create profiles that include their educational background, professional experience, interests, and areas of expertise. Professional information is verified before users are permitted to publish content within their recognized fields.

Content Approval: Submitted content goes through a review process before publication to help ensure that it is relevant, appropriate, and aligned with the contributor's area of expertise.

Expertise-Based Content Sharing: Verified professionals can contribute content within their recognized areas of expertise, helping maintain the reliability and relevance of information shared on the platform.

Premium Content: Professionals can offer premium educational content that users can access through subscription-based features.

Educational Hub: SAMEdu provides an educational content feed where users can discover and learn from material shared by verified professionals.

Functional Requirements

User Registration: Users can create accounts and provide information required for profile and professional verification.

Content Submission: Verified professionals can create and submit educational content for review.

Content Approval: Submitted content can be reviewed, approved, or rejected before publication based on relevance and alignment with the contributor's area of expertise.

Non-functional Requirements

Security: The system should protect user information and maintain the privacy and security of account and profile data.

Performance: The system should provide responsive access to platform features and support efficient content submission and review processes.

Scalability: The system should accommodate growth in users, content, and platform activity without significant degradation in performance.

Additional Requirements

User Feedback: Users can report inappropriate or inaccurate content and provide feedback about the platform.

User Profiles: All users can view, update, and customize their profiles and relevant account information.

Notification System: Users receive notifications about relevant platform activity, including the status of submitted content.

Business Benefits

Enhanced Learning Experience: SAMEdu transforms familiar social media interactions into opportunities for continuous learning and knowledge discovery.

Trustworthy Information: Professional verification and content review help create a community where users can access more reliable and relevant educational information.

Monetization for Professionals: Subscription-based premium content provides professionals with an opportunity to monetize their expertise while contributing educational material to the platform.

Focused and Relevant Content: Expertise-based content sharing helps keep professional content relevant to contributors' verified educational and professional backgrounds.

Users

SAMEdu is designed to support two primary user groups: professionals who contribute knowledge and learners who use the platform to discover educational content.

Professional Users

Professional users have verified educational or professional backgrounds and can contribute content within their recognized areas of expertise. They can share educational material, connect with other professionals, and create premium content for subscribers.

Student/Learner Users

Student and learner users primarily use SAMEdu to discover educational content shared by professionals. They can explore subjects that interest them, interact with educational material, customize their profiles, and connect with the broader learning community.

User Stories

Professional User Story

As a professional, I want to connect with other professionals in my field so that I can expand my network, exchange knowledge, and stay informed about relevant topics and developments.

Acceptance Criteria:

- I can search for professionals based on their industry or area of expertise.
- The platform can suggest relevant professional connections based on my profile and interests.
- I can send connection requests to other professionals.

Student/Learner User Story

As a learner, I want to discover interesting educational content across a variety of subjects so that I can learn in an engaging and accessible way during my free time.

Acceptance Criteria:

- I can discover interesting facts and educational content across different subject areas.
- I can select subjects and topics that interest me.
- The platform can recommend relevant educational content based on my interests.
- Educational content is presented in a clear and engaging format.

Use Cases

Profile Creation: Users create and customize profiles containing information such as their educational background, professional experience, interests, and areas of expertise.

Content Sharing: Verified professionals create and share educational content within their recognized areas of expertise. Learners can discover and interact with published content.

Content Approval: Submitted professional content goes through a review process before publication to help ensure its relevance and alignment with the contributor's area of expertise.

Premium Content Creation: Professionals can create premium educational content that users can access through subscription-based features.

Learning by Scrolling: Users discover educational content through a social-media-style feed designed to make informal learning part of everyday browsing.

Connecting Professionals: Professional users can discover and connect with other professionals based on shared interests, industries, or areas of expertise.

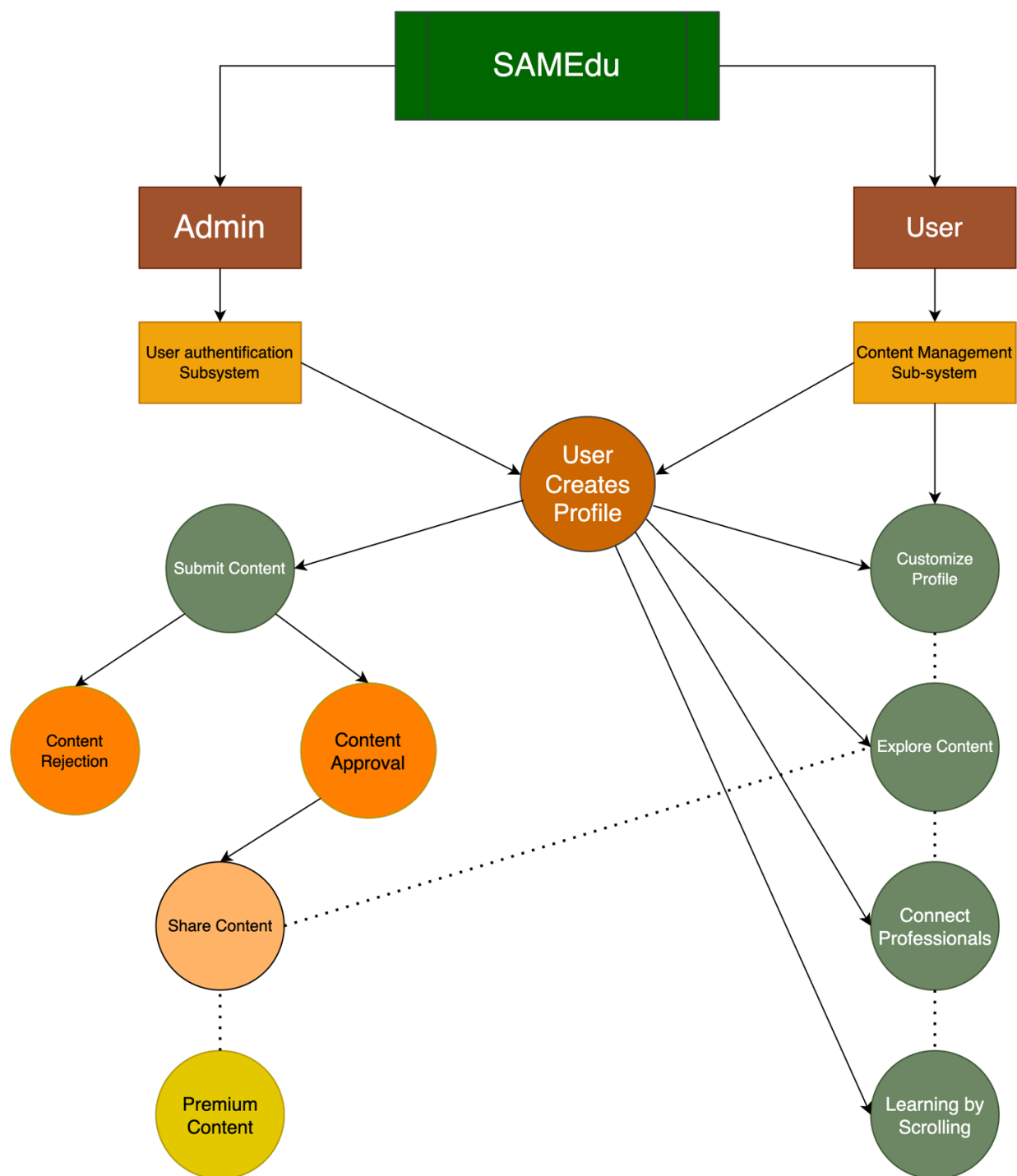
Submit Content: Professional users submit educational content for review before publication. Submitted content is added to the content review process.

Approve Content: Reviewers evaluate submitted content and approve or reject it based on the platform's content requirements.

Explore Content: Users browse and interact with educational content organized around subjects and areas of interest.

Customize Profile: Users update and customize their profile information, interests, and other relevant account details.

Use Case Diagram



Full Use Case Descriptions

Profile Creation

Description: Users create profiles containing information such as educational background, professional experience, interests, and areas of expertise.

Actors: User

Preconditions: The user is logged into the system.

Flow: The user selects the option to create or complete a profile, enters the required information, and submits the profile.

Postconditions: The profile is successfully created and stored in the system.

Exception Conditions: If the user cancels the process or fails to provide required information, the profile is not created.

Content Sharing

Description: Verified professionals create and share educational content within their recognized areas of expertise, while learners browse and interact with published content.

Actors: Professional User, Learner User

Preconditions: The user is logged into the system. Professional users must have a verified profile before submitting professional content.

Flow: A professional user creates a new post, enters the content, selects relevant tags or subject areas, and submits the post for review. Once approved, the content becomes available for learners to view and interact with.

Postconditions: Approved content is published and made available to users on the platform.

Exception Conditions: If the professional user is not verified or the submitted content does not meet platform requirements, the content is not published.

Content Approval

Description: Submitted professional content is reviewed before publication to help ensure relevance and alignment with the contributor's area of expertise.

Actors: Reviewer

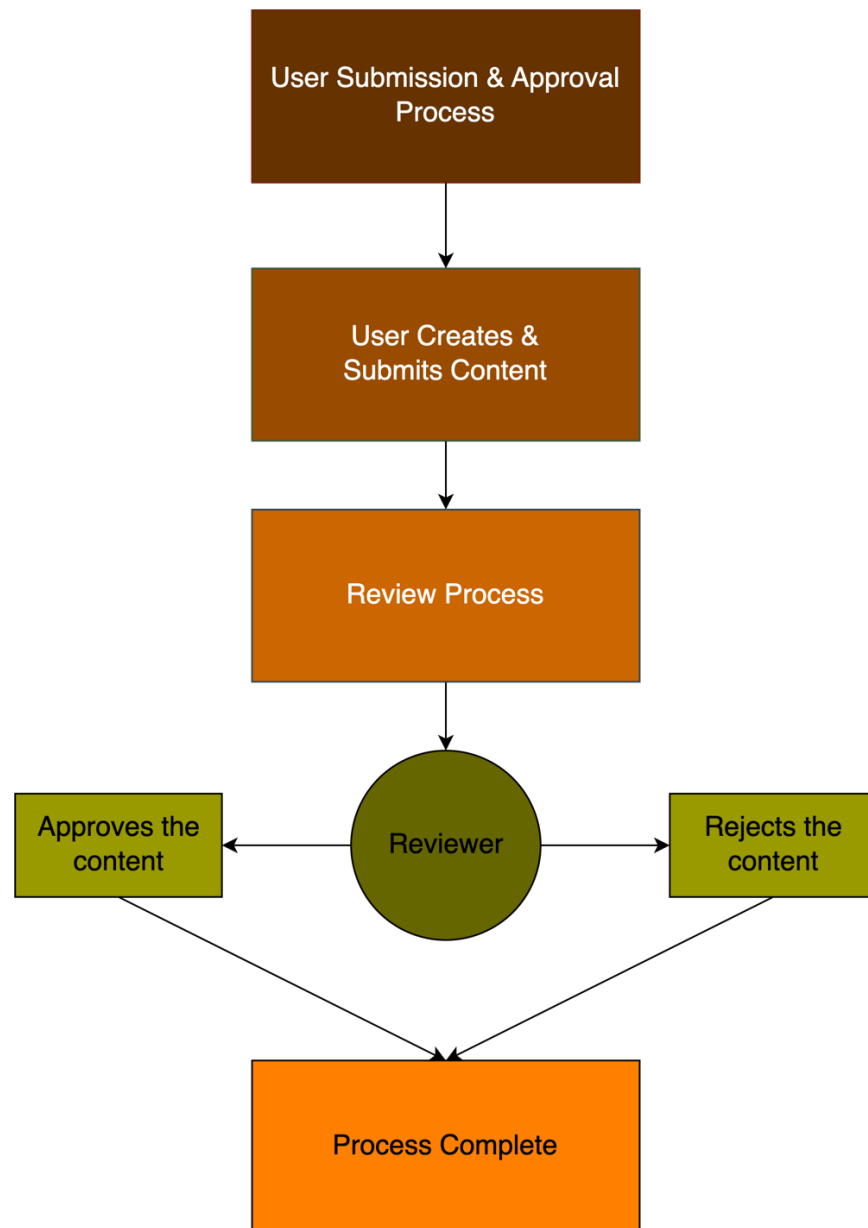
Preconditions: Content has been submitted and is awaiting review.

Flow: The reviewer accesses the queue of pending submissions, selects a submission, evaluates the content, and either approves or rejects it.

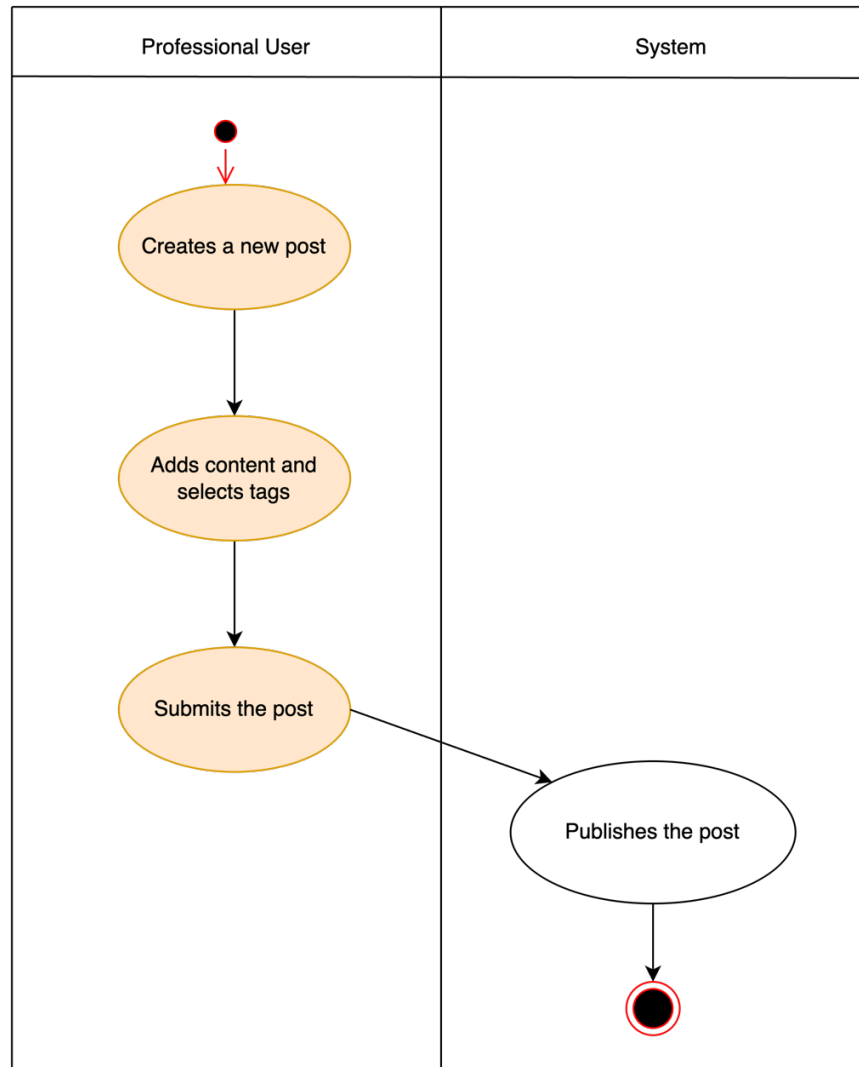
Postconditions: Approved content is published. Rejected content is not published and remains unavailable to other users.

Exception Conditions: If the submitted content is incomplete, inappropriate, or outside the contributor's verified area of expertise, the reviewer may reject the submission.

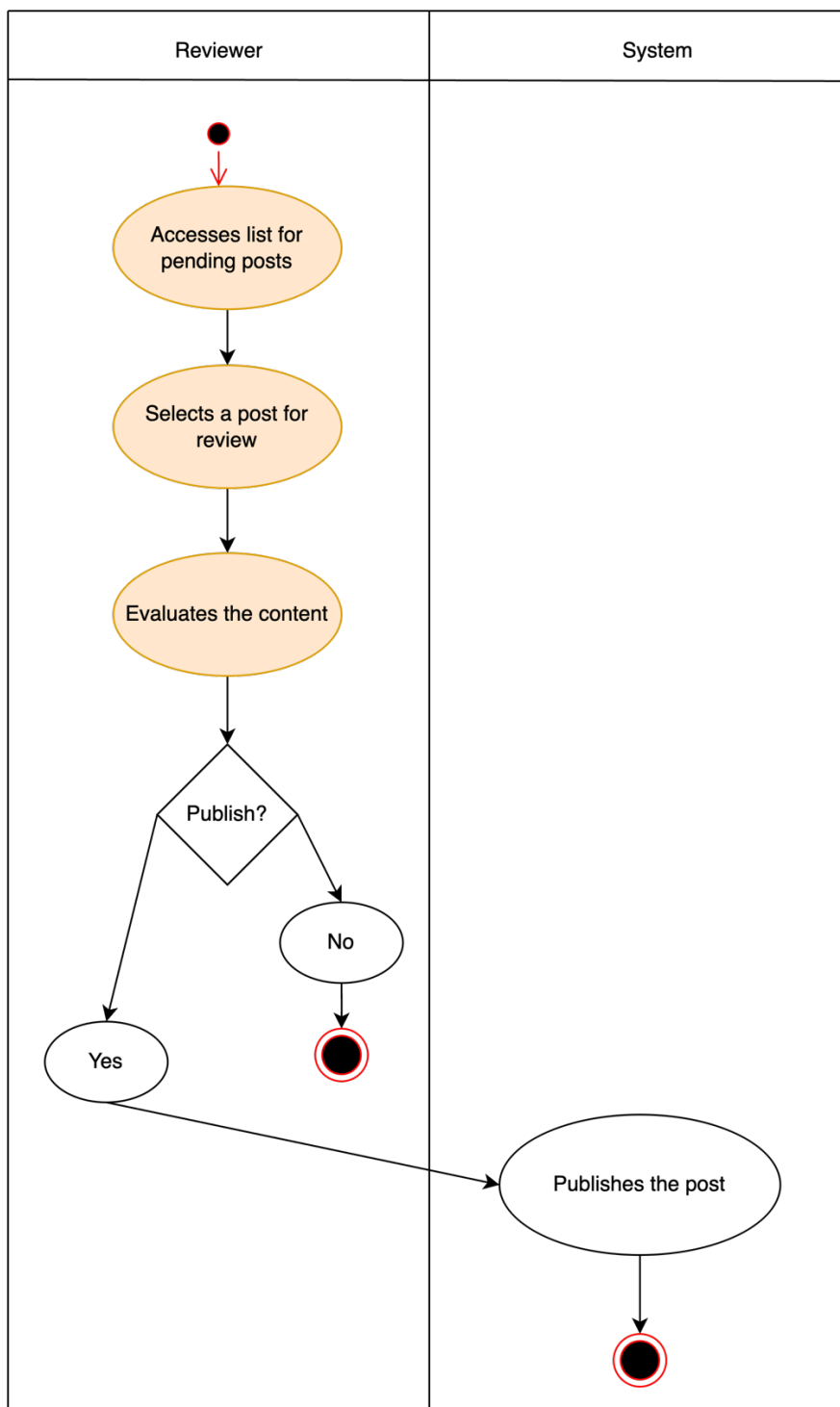
Activity Diagrams



Content Sharing



Content Approval



CRUD Analysis

Use Case \ Domain Class	User	Profile	Content	Premium Content	Review Process	Tag	Category
Profile Creation	C/R	C/R/U	-	-	-	-	-
Content Sharing	C/R	-	C/R/U/D	-	-	C/R	-
Content Approval	-	-	-	-	C/R/U	-	-
Premium Content Creation	C/R	-	C/R/U	C/R/U/D	-	-	-
Learning by Scrolling	C/R	-	C/R	-	-	-	-
Connecting Professionals	C/R	-	-	-	-	-	-
Submit Content	C/R	-	C	-	-	-	-
Approve Content	-	-	-	-	C/R/U	-	-
Explore Content	C/R	-	R	-	-	-	-
Customize Profile	C/R/U	C/R/U	-	-	-	-	-

Classes

User

Attributes: UserID (Key), Name, Educational Background, Professional Experience, Interests.

Post

Attributes: Content, Timestamp, Approval Status.

Subscription

Attributes: User, Premium Content Access, Subscription Period.

Review Process

Attributes: Post, Review Status, Reviewer.

Connection:

Attributes: ConnectionID (Key), User 1, User 2, Shared Interests.

Content

Attributes: ContentID (Key), UserID (Foreign Key), Text, Timestamp, ApprovalStatus.

Reviewer

Attributes: ReviewerID (Key), Name, Expertise.

Profile

Attributes: UserID (Foreign Key), SubscriptionStatus, PremiumContentAccess.

Notification

Attributes: NotificationID (Key), UserID (Foreign Key), ContentID (Foreign Key), Type, Timestamp, Status.

Tag

Attributes: TagID (Key), Name.

Comment

Attributes: CommentID (Key), UserID (Foreign Key), ContentID (Foreign Key), Text, Timestamp.

Category

Attributes: CategoryID (Key), Name.

SubscriptionPlan

Attributes: PlanID (Key), Name, Price, Features.

Message

Attributes: MessageID (Key), SenderID (Foreign Key), ReceiverID (Foreign Key), Text, Timestamp, Status.

Event

Attributes: EventID (Key), UserID (Foreign Key), Title, Description, Date, Location.

Class Connections

User

Associations and Multiplicity

User to Connection: One User can have many Connections.

User to Profile: One User can have at most one Profile.

User to Notification: One User can receive many Notifications.

User to Comment: One User can make many Comments.

User to Message: One User can send and receive many Messages.

User to Event: One User can organize or participate in many Events.

Post

Associations and Multiplicity

Post to Review Process: One Post can undergo a Review Process.

Subscription

Associations and Multiplicity

Subscription to User: One Subscription is associated with one User.

Review Process

Associations and Multiplicity

Review Process to Content: One Review Process is associated with one Content.

Review Process to Reviewer: One Review Process involves one Reviewer.

Connection

Associations and Multiplicity

Connection to User: One Connection involves two Users.

Content

Associations and Multiplicity

Content to Tag: One Content can have many Tags.

Reviewer***Associations and Multiplicity***

Reviewer to Review Process: One Reviewer can be associated with many Review Processes.

Profile***Associations and Multiplicity***

Profile to User: One Profile is associated with one User.

Notification***Associations and Multiplicity:***

Notification to User: One Notification is associated with one User.

Notification to Content: One Notification is associated with one Content.

Tag***Associations and Multiplicity:***

Tag to Content: One Tag can be associated with many Contents.

Comment***Associations and Multiplicity:***

Comment to User: One Comment is made by one User.

Comment to Content: One Comment is associated with one Content.

Category***Associations and Multiplicity:***

Category to Content: One Category can be associated with many Contents.

SubscriptionPlan

Associations and Multiplicity:

SubscriptionPlan to Content: One SubscriptionPlan can be associated with many Contents.

Message***Associations and Multiplicity***

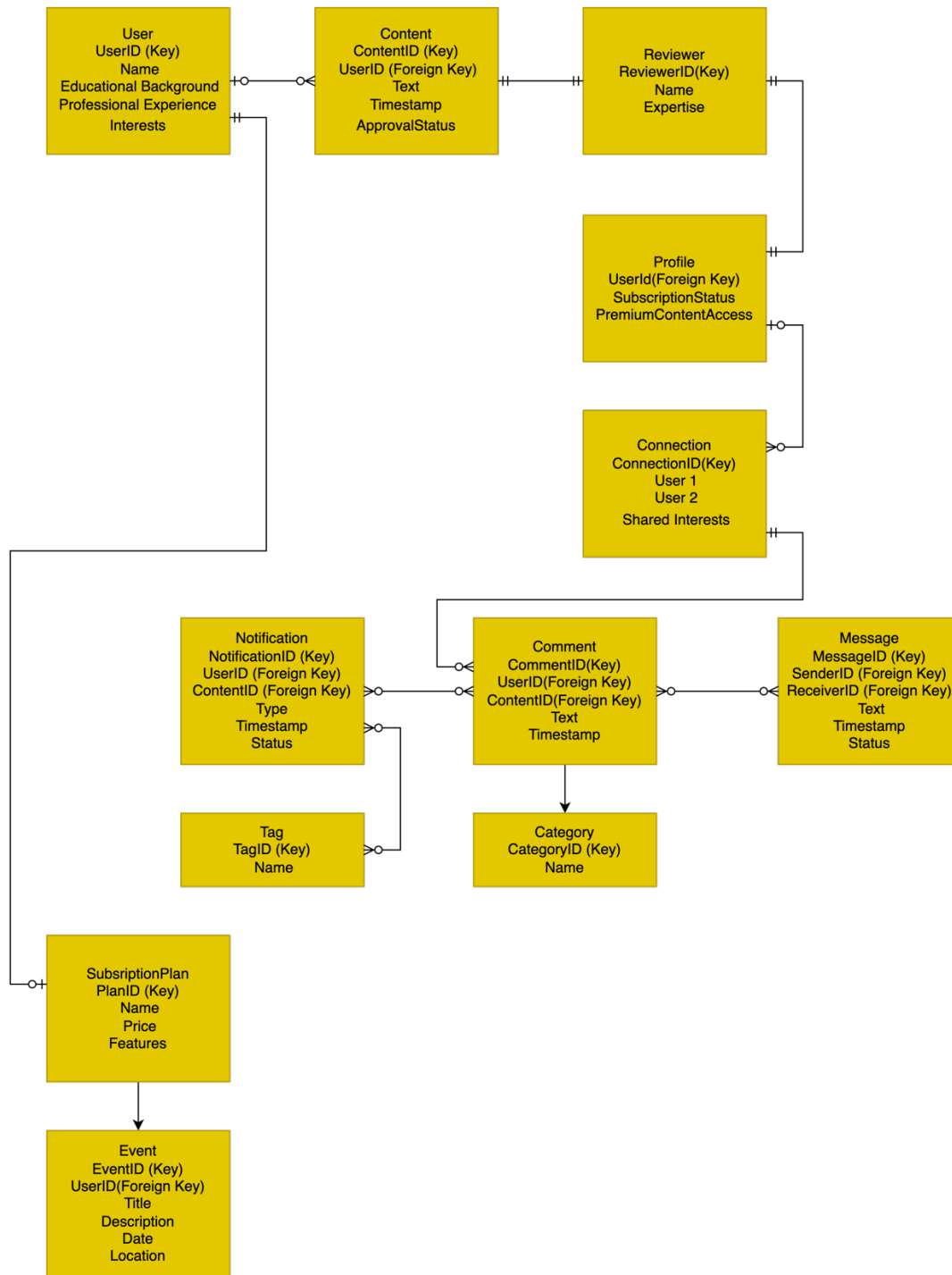
Message to User (Sender): One Message is sent by one User.

Message to User (Receiver): One Message is received by one User.

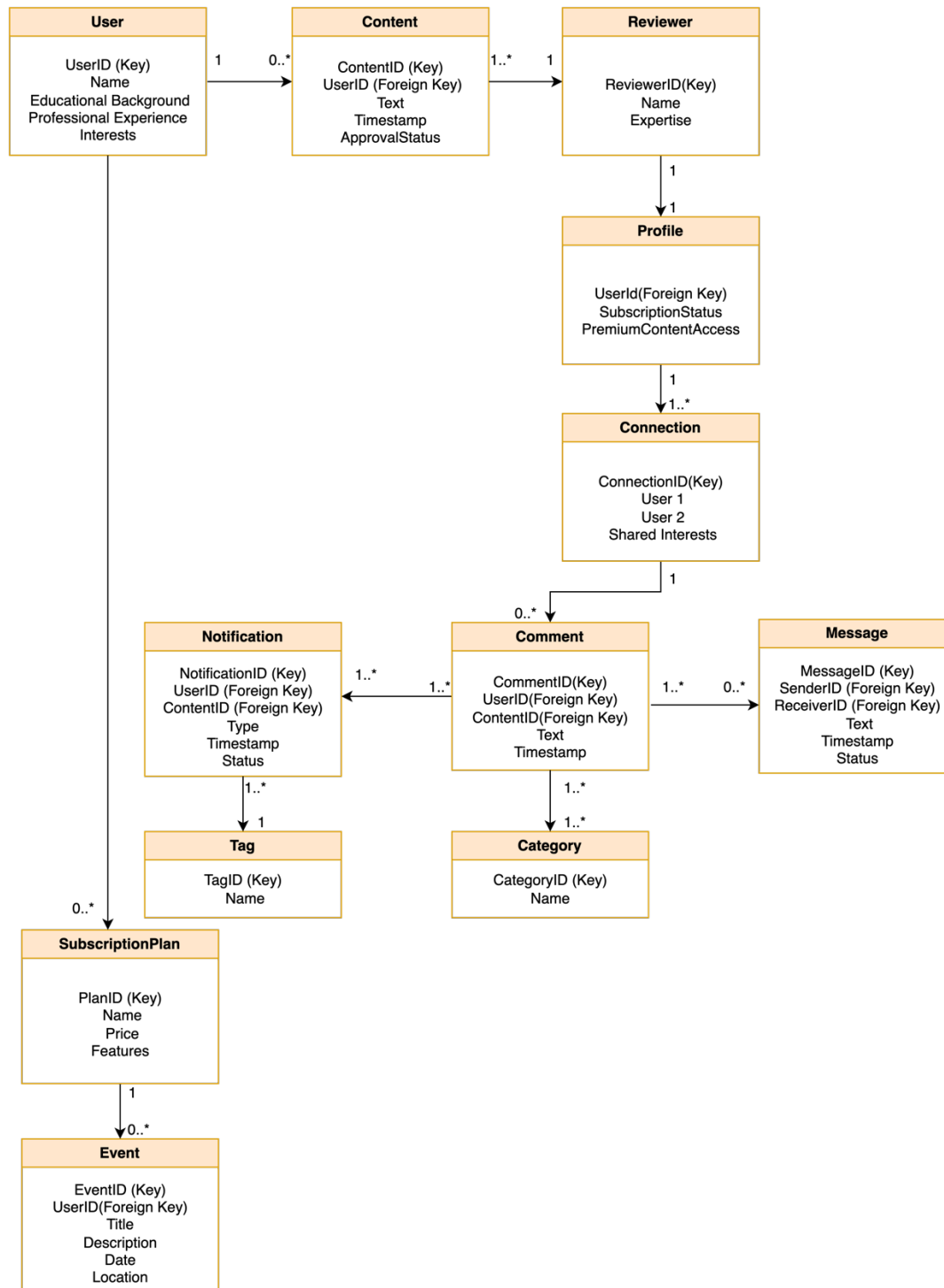
Event***Associations and Multiplicity***

Event to User: One Event is organized by or involves one User.

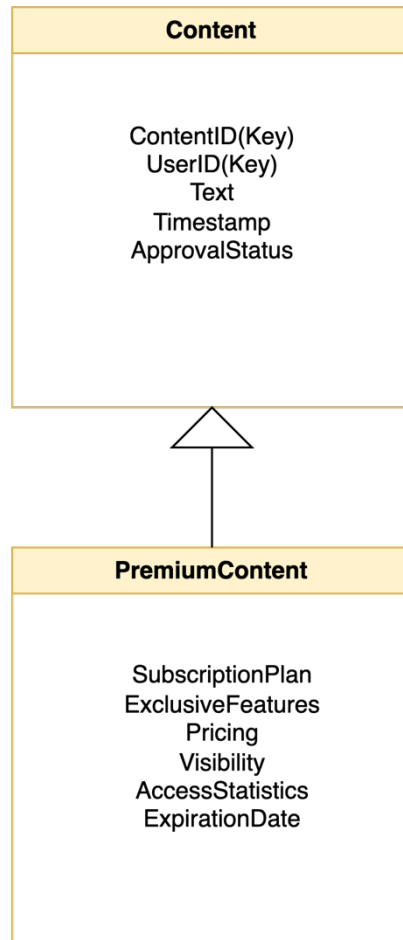
Entity Relationship Diagram (ERD)



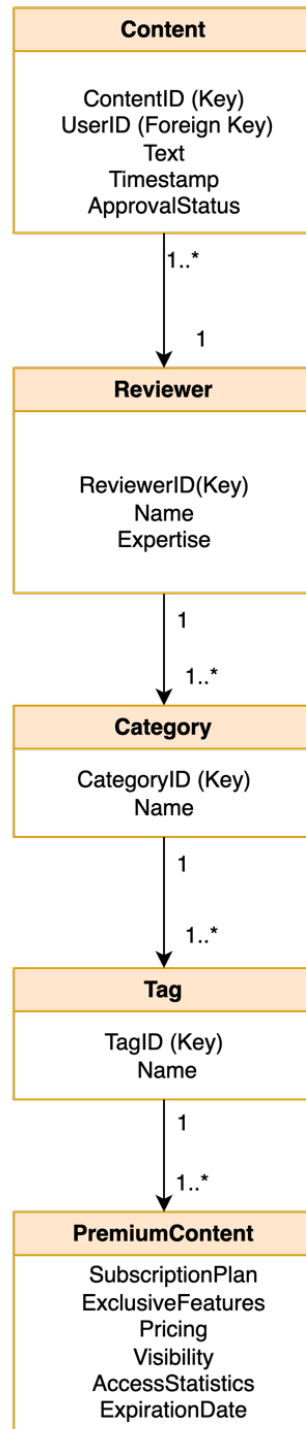
Domain Class Diagram



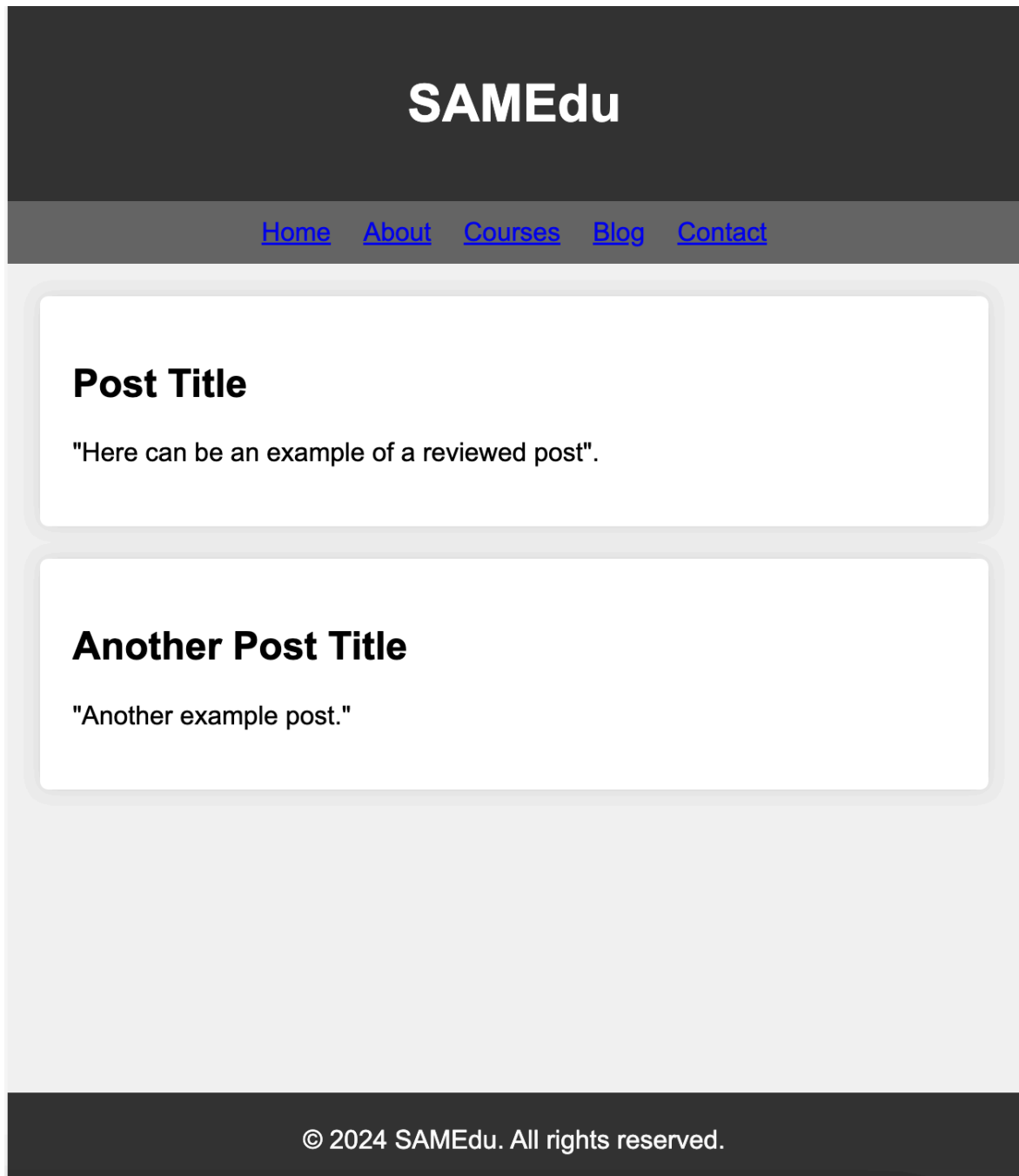
Inheritance Diagram



Content Management Subsystem Diagram



Interface Design



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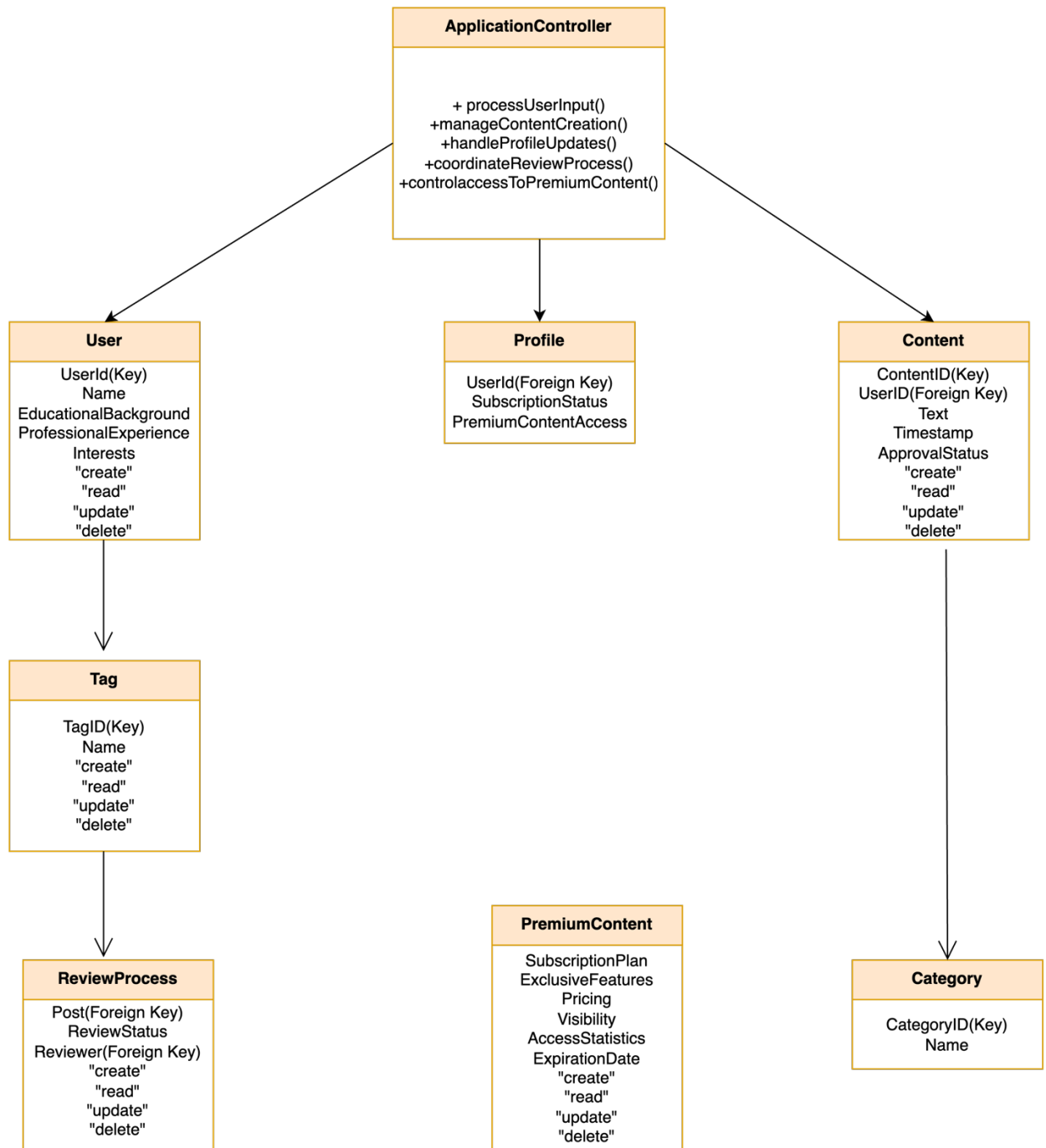
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Controller Class Diagram



CRC Cards

Card1

User (Class Name)	
(Responsibilities) • Update Profile • Share Content • Submit Content for Review • Explore Educational Content	(Collaborating Classes) • Profile • Content • Review Process

UserID Name EducationalBackground ProfessionalExperience Interests
--

Card2

Profile (Class Name)	
(Responsibilities) • Create Profile • Customize Profile • View Subscription Status	(Collaborating Classes) • User • Subscription

UserID SubscriptionStatus PremiumContentAccess
--

Communication Diagrams

Diagram 1: Profile Creation

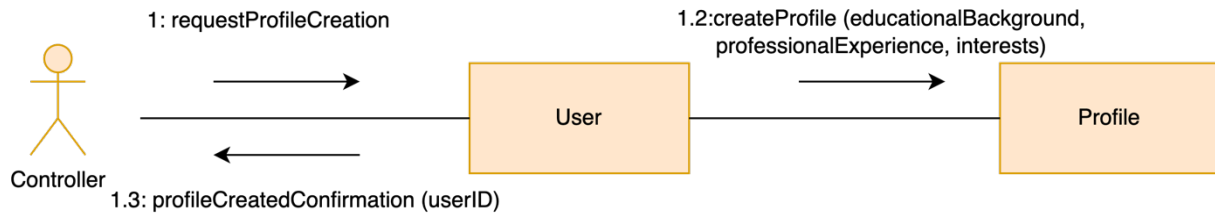


Diagram 2: Content Sharing

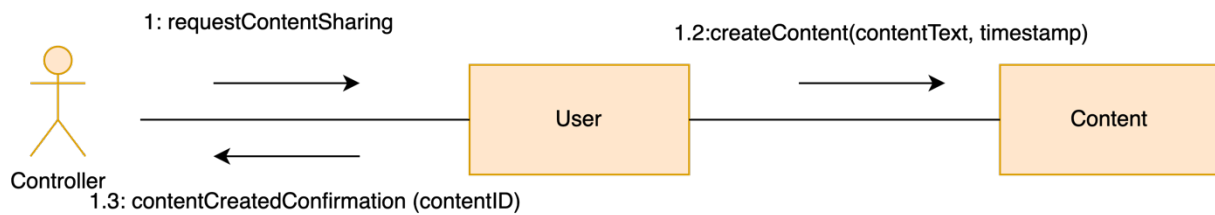
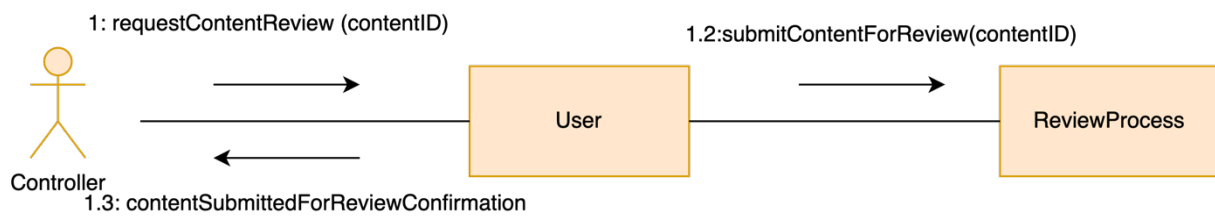
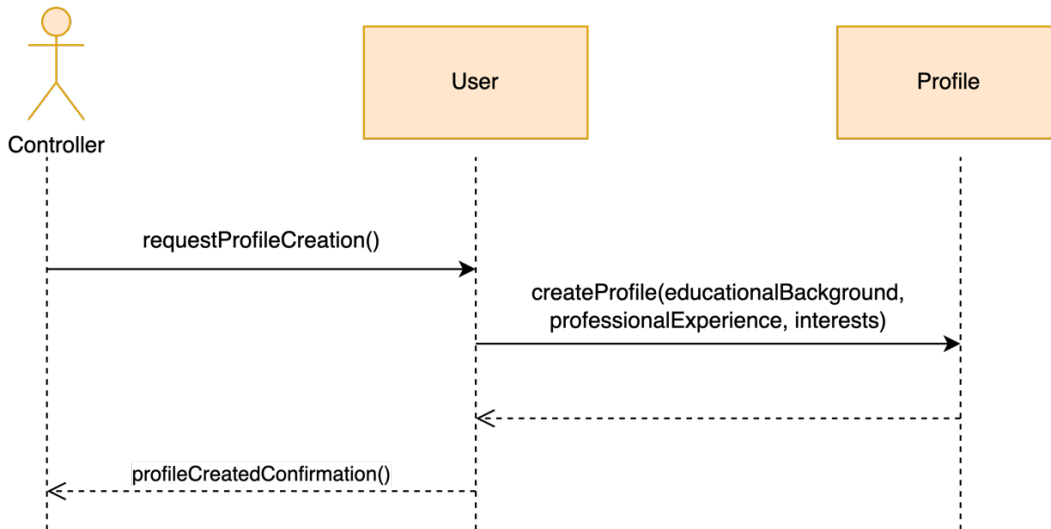


Diagram 3: Content Review

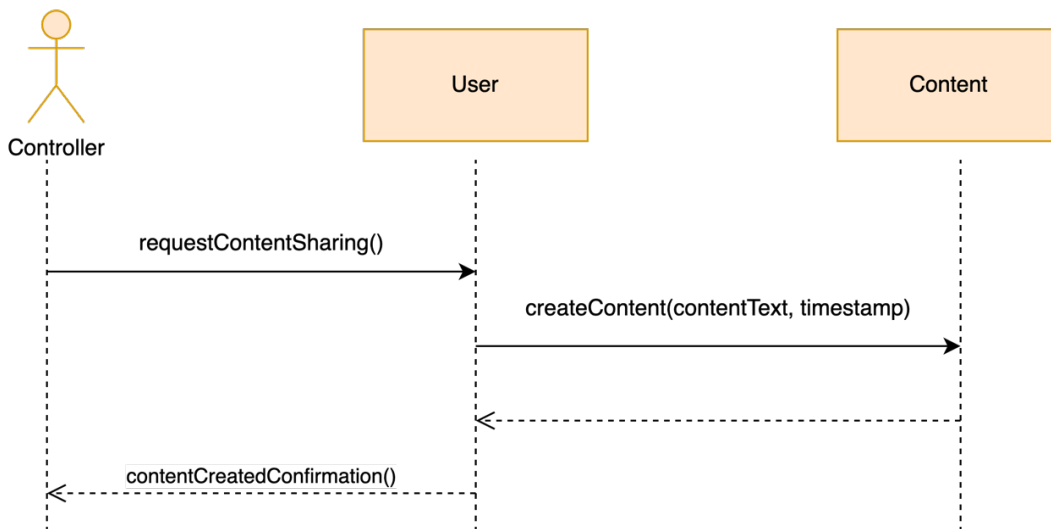


Sequence Diagrams

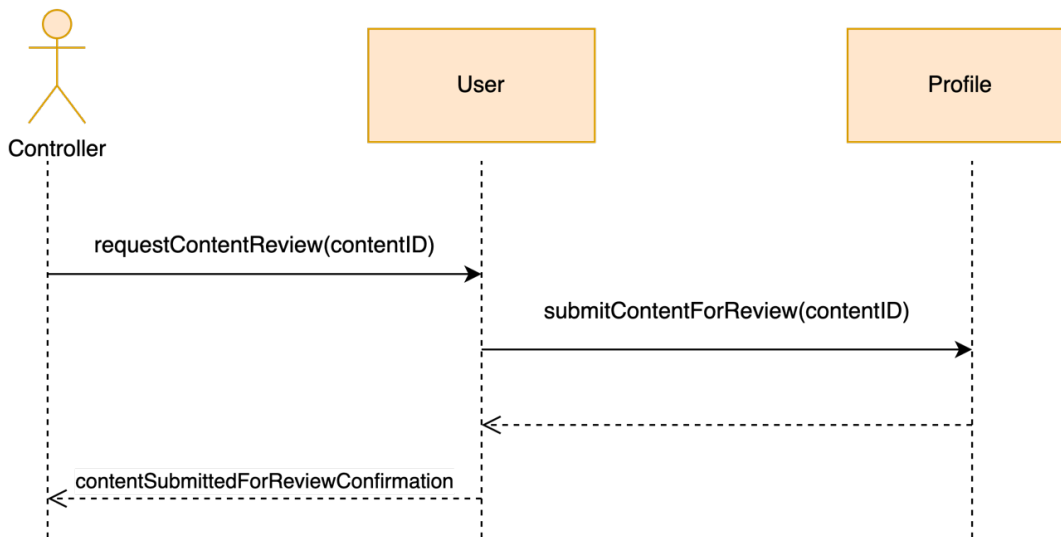
Profile Creation



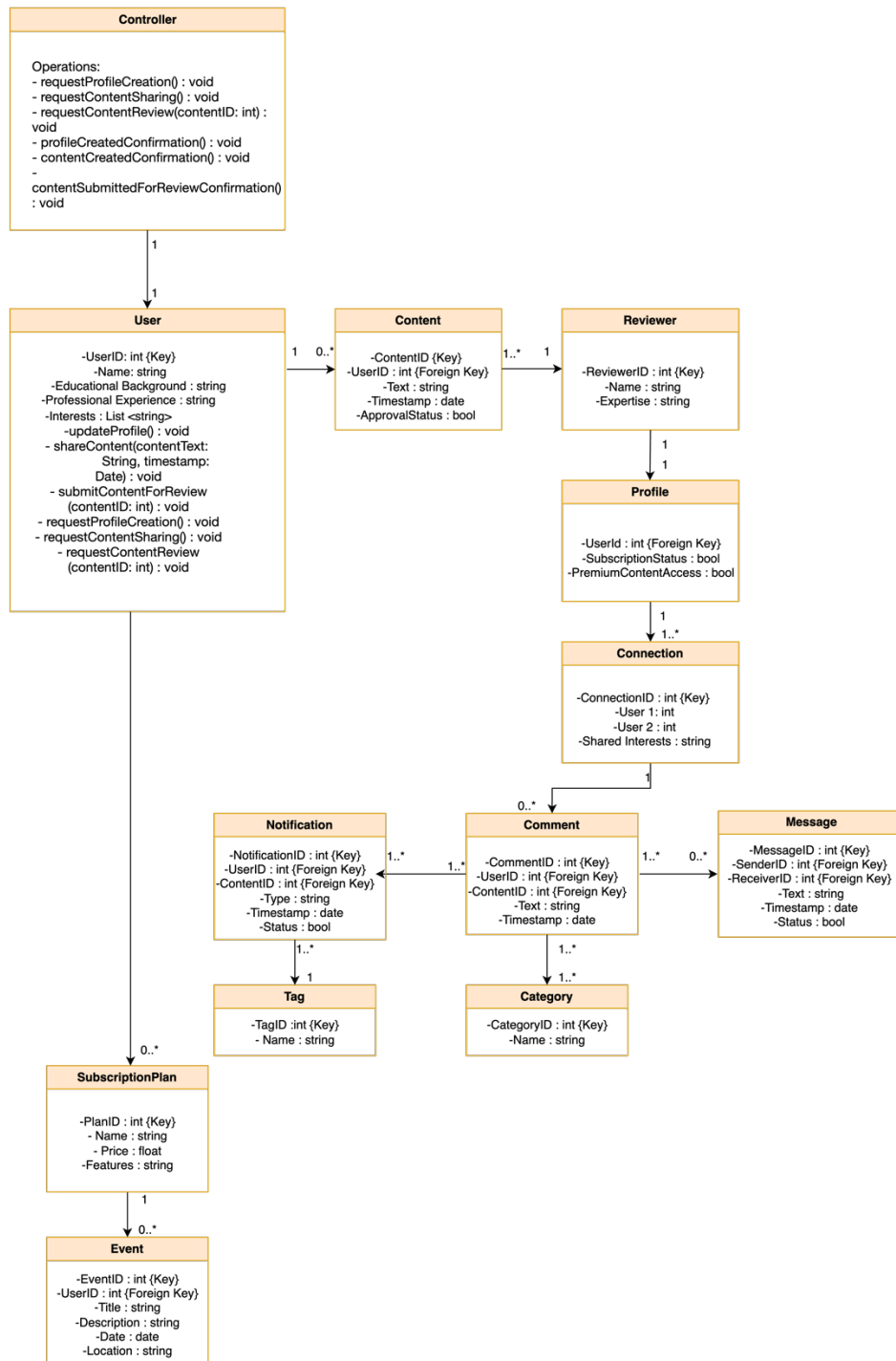
Content Sharing



Submit Content for Review



Class Diagram



User Interface Design

One of the use cases that we identified earlier is Profile Creation. It had three attributes: Educational Background, Professional Experience, and Interests. The following mockup illustrates a possible user interface for this feature.

The mockup shows a white card with rounded corners on a light gray background. At the top, the title 'SAMEdu Platform - Profile Creation' is displayed in a large, bold, black font. Below the title, there are three sections, each with a bold label and a corresponding text input field. The first section is 'Educational Background:' with a text box containing the placeholder 'Enter your educational background'. The second section is 'Professional Experience:' with a text box containing the placeholder 'Enter your professional experience'. The third section is 'Interests:' with a text box containing the placeholder 'Enter your interests'. At the bottom left of the card, there is a green button with the text 'Create Profile' in white.

The following mockup illustrates a possible login interface for the platform.

The mockup shows a dark gray header bar with the text 'SAMEdu Platform' in white. Below the header, there is a white card with rounded corners. Inside the card, the text 'Welcome to SAMEdu Platform! Please login to access your account.' is displayed. Below this text, there is a blue button with the text 'Login' in white.

FAQ

Q: Why was SAMEdu designed?

A: SAMEdu was designed to combine social media with educational content and knowledge sharing. The platform aims to create an environment where professionals can share their expertise and learners can discover reliable educational content through familiar social media interactions.

Q: Who can benefit from using SAMEdu?

A: SAMEdu is designed for professionals, educators, students, and other learners. Professionals can contribute knowledge within their areas of expertise, while learners can explore educational content based on their interests.

Q: What sets SAMEdu apart from traditional social media platforms?

A: SAMEdu focuses specifically on educational and professional knowledge sharing. Professional verification, expertise-based content sharing, and content review are designed to improve the relevance and reliability of information published on the platform.

Q: How does SAMEdu support trustworthy content?

A: Users who contribute professional content provide information about their educational and professional backgrounds as part of the verification process. Submitted content also goes through a review process before publication to help ensure that it is relevant and aligned with the contributor's area of expertise.

Q: How does SAMEdu make learning more accessible?

A: SAMEdu presents educational content through a familiar social-media-style experience.

Users can browse subjects that interest them, discover content shared by professionals, and incorporate informal learning into everyday browsing.

Q: How does SAMEdu address misinformation?

A: SAMEdu uses professional verification, expertise-based content sharing, and content review to reduce the likelihood of irrelevant or unreliable professional information being published.

These mechanisms are designed to create a more focused environment for educational knowledge sharing.